COMPUTER NETWORKS
PHASE 1 – LITERATURE REVIEW AND RESEARCH GAP IDENTIFICATION

Proposed Research Topic:
Graph Neural Networks for Topology-Aware AIOps: Network Anomaly Detection and Root-Cause Localization

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Critical Summary

Paper 1 – Luo et al., "Topology-aware self-supervised multivariate KPI anomaly detection and root cause node localization for network operations." [1]

The paper studies anomaly detection and root-cause localization from multivariate network KPIs. It uses a Mininet testbed with three topology scales and six fault categories, and also validates on the public Train-Ticket benchmark. The method combines masked reconstruction, one-step prediction and topology-aware representation learning, then ranks root-cause nodes using local anomaly evidence and directed propagation. The reported Mininet results are an F1-score of 0.9235 and A@1 of 0.7613; external validation gives F1 0.846 and A@1 0.742. The authors note that the F1 improvement over DCdetector is small and not statistically significant, and that the comparison with Nezha uses a restricted KPI-plus-topology input setting. A practical limitation is that most experiments are still based on an emulated environment with a limited number of fault types. Generalization to much larger or rapidly changing production topologies is not established. The study also stops at detection and localization; failure prediction before the fault and automated remediation are not evaluated.

Paper 2 – Zhou et al., "Root cause analysis of cloud platform faults based on anomaly correlation graph and graph neural network." [2]

This work focuses on root-cause analysis in cloud platforms where one fault can trigger many dependent and noisy alarms. Historical alarm sequences and component topology are used to build an Anomaly Correlation Graph through a time-delay topological Hawkes process. A weight-threshold sampling step removes weakly related neighbors, a Graph Attention Network extracts graph features, and LightGBM performs the final classification. The paper reports more than a 10% improvement in precision over relevant comparison methods. The authors explicitly limit the study to supervised learning from historical data because root-cause alarms are treated mainly as recurring failure modes. This makes the method less convincing for previously unseen faults. The model also depends on the quality and completeness of alarm data and on the learned correlation structure. Missing experiments include cross-topology transfer, proactive failure prediction and closed-loop recovery after the root cause is identified.

Paper 3 – Cressant et al., "On Graph Design for GNN-Based Network Anomaly Detection." [3]

This paper addresses a basic GNN design problem: network telemetry is multivariate time-series data, but the graph edges are not always known. The proposed COSI method constructs graphs by combining semantic similarity between feature names with statistical correlation between time series. It is evaluated on three datasets, including two real-world networking datasets and one benchmark dataset, and improves GNN anomaly-detection performance over conventional graph construction approaches on almost all reported evaluation scores. The main strength is that graph construction is treated as part of the learning problem rather than as a fixed assumption. A limitation is that correlation does not necessarily represent a physical or causal dependency. Semantic similarity can also be sensitive to vendor-specific KPI naming. The paper mainly studies anomaly detection rather than root-cause localization or prediction. A useful next step is therefore to test dynamic or hybrid graphs that combine known physical topology with learned dependencies and to evaluate them when the network topology changes.

Cross-Paper Comparison Table

#	Title	Year	Dataset Used	Preprocessing Techniques	Models / Techniques Used	Evaluation Metrics Used	Results	Limitations / Research Gap
1	Topology-aware self-supervised multivariate KPI anomaly detection and root cause node localization for network operations	2026	Mininet testbed: three topology scales, six fault categories; external Train-Ticket benchmark	KPI windowing; masked reconstruction; one-step prediction; topology-aware representation; residual-based anomaly scoring	Self-supervised temporal learning with topology regularization and directed multi-hop root-cause ranking	F1-score; A@1; multi-seed stability; runtime	Mininet F1 0.9235; A@1 0.7613. Train-Ticket F1 0.846; A@1 0.742	Mostly controlled/emulated data; limited fault set; modest topology scale; no proactive prediction or remediation; cross-topology generalization remains open
2	Root cause analysis of cloud platform faults based on anomaly correlation graph and graph neural network	2026	Historical cloud-platform alarm-event sequences with component topology	Time-delay modeling; anomaly correlation graph construction; weight-threshold neighbor sampling	Time-delay Topological Hawkes Process, GAT, LightGBM, historical-case similarity analysis	Precision and root-cause localization comparisons; ablation analysis	Precision improved by more than 10% over relevant comparison methods	Supervised recurring-fault setting; relies on historical alarms and learned correlations; unseen-fault and cross-topology behavior not established
3	On Graph Design for GNN-Based Network Anomaly Detection	2026	Three datasets: two real-world networking datasets and one benchmark dataset	Feature-name semantic similarity plus statistical time-series correlation to construct the graph	COSI graph construction used with GNN-based anomaly detection models	Multiple anomaly-detection evaluation scores across the three datasets	COSI improves performance over conventional graph designs on almost all reported scores	Correlation is not causation; feature naming may be vendor-dependent; focuses on detection rather than root cause, prediction or recovery; graph drift remains open

Research Gap Synthesis

Research Gap	Evidence from Paper 1	Evidence from Paper 2	Evidence from Paper 3	Proposed Research Opportunity
Gap 1: Cross-topology generalization	Uses multiple Mininet scales, but evaluation is still mainly within controlled topologies.	Learns from historical alarm and topology relationships in a cloud platform; transfer to a different topology is not shown.	Shows that graph design strongly affects GNN anomaly-detection performance.	Train on several network topologies and test on a previously unseen topology. This would measure whether the GNN learns transferable failure patterns rather than memorizing one graph.
Gap 2: Static graph assumptions in changing networks	Uses known topology as an important prior.	Builds an anomaly-correlation graph from historical alarm relationships.	COSI creates graph edges from semantic similarity and statistical correlation.	Develop a dynamic or hybrid graph that combines physical links with learned statistical dependencies and updates them as traffic, routing or service dependencies change.
Gap 3: Reactive diagnosis instead of proactive failure prediction	Detects anomalies and ranks root-cause nodes after abnormal behavior is present.	Focuses on locating the root cause of a fault from alarms.	Focuses on anomaly detection and graph construction.	Use a spatio-temporal GNN to predict whether a node or link is likely to fail within a future time window, while retaining topology information for localization.
Gap 4: Limited access to large labelled operational datasets	Uses an emulated Mininet environment and a public benchmark because labelled operational faults are difficult to obtain.	Depends on historical cloud fault and alarm data that may not be available publicly in other environments.	Uses available datasets, but the broader problem of creating useful graph structures from operational telemetry remains.	Generate a controlled dataset with Mininet or a similar emulator by running multiple topologies, collecting telemetry and injecting labelled faults such as congestion, packet loss, link failure and routing disturbances.

References

[1] D. Luo, W. Wang, X. Liu, W. Yang, K. Hu, et al., "Topology-aware self-supervised multivariate KPI anomaly detection and root cause node localization for network operations," *Scientific Reports*, 2026. doi: 10.1038/s41598-026-69392-8.

[2] B. Zhou, B. Chen, L. Yan, X. Liang, and Z. M. Ma, "Root cause analysis of cloud platform faults based on anomaly correlation graph and graph neural network," *Engineering Applications of Artificial Intelligence*, vol. 174, Art. no. 114451, 2026. doi: 10.1016/j.engappai.2026.114451.

[3] K. Cressant, F. Larroca, S. Secci, and P. Braconnot Velloso, "On Graph Design for GNN-Based Network Anomaly Detection," *IEEE Transactions on Network and Service Management*, vol. 23, pp. 4122–4134, 2026. doi: 10.1109/TNSM.2026.3684653.